

Intrusion Detection in IoT Networks using LSTM Deep Learning Models with the UNSW-NB15 Dataset

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Abstract— Network Intrusion Detection Systems (IDS) are an essential part of the Internet of Things (IoT) Networks, addressing the increasingly severe challenges posed by Internet security threats. This research enhances (IDS) capabilities in IoT environments by investigating the performance of Deep Learning Models with Binary Classification for identify and predict if the connection is normal or attack. The Deep Learning Models; including Long Short-Term Memory (LSTM), Bidirectional LSTM, Gated Recurrent Unit (GRU) and Bidirectional GRU. The comprehensive UNSW-NB15 Dataset is utilized that includes real-world network traffic and a variety of attacks. To effectively tackle the issue of Class Imbalance in the Dataset, we utilize the Synthetic Minority Over-sampling Technique (SMOTE). This approach enhances the representation of minority classes, leading enhances our model's performance and robustness. We train and evaluate these models, our evaluation based on Accuracy, Recall, Precision, F1-score and AUC metrics. The results indicate that the model with the best metrics is the Bidirectional LSTM BiLSTM, achieving an Accuracy of 0.9468 , AUC of 0.9801 without SMOTE and an Accuracy of 0.9501 , AUC of 0.9953 with SMOTE. These findings emphasize the effectiveness of the BiLSTM model for a more accurate and robust Intrusion Detection Systems (IDS) using the SMOTE technique and underscore the beneficial impact of the SMOTE technique on class imbalance issues. The research contributes valuable insights towards the development of more robust and secure IoT systems using the LSTM model using the SMOTE technique.

Keywords— *Intrusion Detection, UNSW-NB15 Dataset, SMOTE-Oversampling, Deep Learning, LSTM, BiLSTM, GRU, BiGRU*

I. INTRODUCTION

Today, many devices, including smartphones, are connected to the Internet and creating concept Internet of Things (IoT). Internet of Things is a concept that allows devices to connect to the internet and allow devices transmit data automatically without the need for human. The concept "Things" in (IoT) refers to objects that can communicate either with or without human involvement. (IoT) is structured into three layers: perception layer, network layer and application layer [1], IoT systems remain unsafe due exploitation by malicious attacks. Each layer within the IoT architecture presents unique security weaknesses that can be exploited to compromise the entire system [2]. Common IoT attacks include Malicious Operation, Denial of Service, Malicious Control. IoT facilitates the distribution and sharing of vast amounts of information. However sensitive

Information can be accessed and stolen by unauthorized individuals.

To overcome this risk, robust security measures, such as firewalls, are crucial for safe confidential data. A firewall safeguards against external attacks by managing the incoming and outgoing data packets that access the internal system through packet filtering. However, fully securing an information system can be challenging due to the complexity of network configurations and the variety of modern attack methods that may require specific packets to be allowed or denied [3]. To address this issue, Intrusion Detection Systems (IDS) were introduced to detect attacks through depth packet analysis, an IDS can be either hardware or software. The primary function of an IDS is to establish a security layer that alerts administrators to suspicious activity. This is achieved by continuously monitoring network traffic from multiple IDS sensors, enabling the system to identify and report potential threats based on detailed packet inspection [4].

Intrusion Detection Systems (IDSs) face several challenges in effectively achieving their security goals. One significant challenge is the slow detection speed, particularly for signature-based IDSs that rely on matching observed patterns to known misuse patterns. This delay can allow attacks to progress further before being identified. Another critical challenge is the inability of most prevalent IDSs to automatically detect attacks. These attacks, which exploit previously unknown vulnerabilities, are becoming increasingly frequent and sophisticated [5]. The rising number and security of attacks motivate to explore new methods to enhance detection capabilities, ensuring faster, more reliable, and adaptable security measures.

To develop an accurate and high-performance model for identifying Intrusion Detection Systems (IDSs), Machine Learning algorithms can be used to construct prediction model. Arthur Samuel defines machine learning as a field of study that enables machines to learn without being explicitly programmed. Machine learning (ML) handle data more efficiently [6]. The prediction model aims for predicting if the connection is normal or attack then detecting attacks on IoT systems, the objective is to develop a model utilizing machine learning algorithms that can enhance the security of IoT systems and defend against potential attacks.

Researchers use different Machine Learning classifiers for achieving high accuracy model. In recent years the studies tend to use deep learning, Deep learning, a subset of machine learning, has emerged as a highly active research area in

recent years, it enables artificial neural networks to autonomously extract features from data without explicit human supervision. By incorporating advanced training methods and innovative architectural components, modern neural networks can effectively process various data types, including text, tabular data and images [7]. In deep learning, the term “deep” comes from neural networks, which are typically composed of multiple layers in deep learning models [8]. Most popular algorithms deep learning is Recurrent Neural Network (RNN) and it have many types; Long Short-Term Memory (LSTM), Bidirectional LSTM (BiLSTM) , Gated Recurrent Unit (GRU) and Bidirectional GRU (BiGRU).

In this research, we will utilize the UNSW-NB15 Network Intrusion Dataset, which was created at the University of New South Wales Cyber Security Lab and published in 2015. The UNSW-NB15 dataset includes real and modern normal activities and attack behaviors. For our training dataset, we will use the UNSW-NB15 training, which contains 175,341 records and 44 features categorized into nine different types of malicious activities: Denial of Service, Shellcode, Analysis, Backdoor, Exploit, Fuzzers, Generic, Reconnaissance and Worms, in addition to normal traffic. This dataset is available for free use in academic research [9]. The dataset has two classes: normal and attack. Data imbalance is challenge in the UNSW-NB15 dataset such that occurs when the number of samples in class is higher than those in other class. Machine learning on such datasets may yield unsatisfactory results [14]. In the dataset we have Binary classification with normal class and the attack class. There are many methods for solving Data imbalance called under-sampling and over-sampling like random over-sampling (ROS) and Synthetic Minority Over-sampling Technique (SMOTE).

We aim to enhance the quality of the dataset by solving class imbalance using SMOTE method, and we aim to study the performance the four techniques in recurrent neural network (RNN) without SMOTE and with SMOTE. The four models are: Long Short-Term Memory (LSTM), Bidirectional LSTM (BiLSTM), Gated Recurrent Unit (GRU) and Bidirectional GRU (BiGRU). After applying LSTM deep learning models on the UNSW-NB15 dataset. The performance Results will be analyzed to identify the LSTM model that yields the highest performance for this UNSW-NB15 dataset. The remaining research is structured as follows. Section II provides related works. Section III overviews the background of the LSTM models used in this research. Section IV presents the primary methodology is used and describes the experimental steps, section V discusses the results and evaluation metrics, while Section VI provides the conclusion and future work of the entire research.

II. RELATED WORKS

Intrusion prediction has been performed in the literature using various techniques including Machine learning and deep learning techniques. These techniques help to identify, predict attacks.

A. Traditional ML-based IDS:

The authors in [11] compares KDDCUP99 with the UNSW-NB15 dataset to evaluate NIDS performance. Because the KDDCUP99 dataset has been widely used in NIDS research, and it not fully represent contemporary threats. Classification algorithms are investigated: Logistic Regression, Random Forest and Support Vector Machine. Model performance is assessed using Accuracy, Precision, Recall and F1-Score. Results demonstrate that Random Forest with 98.68% accuracy generally outperforms the other methods, aligning with findings from previous studies.

The UNSW-NB15 datasets are imbalanced, with unequal class sample sizes. Machine learning on such datasets may yield unsatisfactory results, prompting the authors in [14] and use of Conditional Tabular Generative Adversarial Networks (CTGAN) to generate synthetic data and balance imbalanced intrusion detection datasets. The authors focus is on multi-class classification, they evaluate the impact of CTGAN-generated synthetic data on the performance of various ML classifiers using the NSL-KDD and UNSW-NB15 datasets. The findings demonstrate significant performance improvements. Compared to training on imbalanced datasets, training on datasets balanced with CTGAN-generated synthetic data resulted in an accuracy increase of up to 8% and an MCC score improvement of up to 13%. Furthermore, this approach consistently outperforms several state-of-the-art intrusion detection systems on both datasets. Also the authors highlight the potential limitations of random oversampling. In some cases, training on randomly oversampled datasets can actually degrade the performance of certain ML classifiers compared to training on the original imbalanced data.

B. Deep Learning based- IDS:

The deep learning techniques are wide used in Intrusion prediction, the authors in [12] compared the performance of Deep Learning algorithms, namely Deep Neural Networks (DNN), Convolutional Neural Networks (CNN), Long Short-Term Memory networks (LSTM), and Auto encoders (AE), in effectively detecting and preventing network intrusions. Using the UNSW-NB15 dataset and for evaluation the binary classification is employed, the research found that the DNN algorithm achieved superior performance with an accuracy of 99.76% and a minimal loss of 0.006%. This finding underscores the critical role of intrusion detection systems in safeguarding IoT devices and networks, and demonstrates the effectiveness of the DNN algorithm in addressing network security breaches within the IoT domain, but the UNSW-NB15 datasets are imbalanced that means with unequal class sample sizes and the authors haven't handling imbalanced datasets so that the Deep Learning algorithms using imbalance datasets may yield unsatisfactory results [14].

LSTM is a deep leaning technique and is an artificial Recurrent Neural Network (RNN) architecture and used in intrusion prediction, the authors in [3] introduce a novel model for predicting the timing of attacks. The model provide Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) architectures within an intrusion detection system (IDS) framework. To enhance model accuracy, the study employs a finite state machine concept to transform

floating-point values in the UNSW-NB15 dataset into equivalent binary representations. The results demonstrate a significant improvement in prediction accuracy. Specifically, the GRU and LSTM models exhibit an average increase in weighted F1 score of 13% and 18%, respectively, when utilizing this binary data representation.

While the authors in [13] proposes an improved network IDS using a deep learning approach. Four deep neural network models were trained with Multi-class classification, including two feed-forward dense networks and recurrent neural networks utilizing Long Short-Term Memory (LSTM) layers. The evaluation was conducted on the UNSW-NB15 dataset, z-score normalization was used for preprocessing the data and cross-validation is used to train the neural network, following data integrity checks and appropriate preprocessing for neural network training. Dense models achieved a validation accuracy of 78.94% without and 79.34% with weight balancing. LSTM models attained 76.84% and 79.21% accuracy, respectively. While both model types benefited from weight balancing, the dense-layer model exhibited slightly higher accuracy. A detailed analysis of confusion matrices revealed significant differences in the models' prediction correctness for different attack categories.

C. Hybrid Approach:

The authors in [10] used the UNSW-NB15 dataset to evaluate the effectiveness of a proposed strategy. The design achieved an accuracy rate of 91.29% by developing a hybrid approach. The activation mechanisms and multiple layers of the deep learning network produce non-convex forms. Adjusting the hyper parameters of deep neural networks may be necessary to reduce the error rate and improve the model's performance. Another challenge is the time required to fine-tune these hyper parameters during training, as well as the categorization metrics of the Network Intrusion Detection System (NIDS).

TABLE I.SUMMARY OF PREVIOUS STUDIES

Reference	Dataset	Model	Key Metrics	Results	Notes
[3]	UNSW-NB15	LSTM GRU	weighted F1	+13%	Predicting Timing of Attacks
[10]	UNSW-NB15	Hybrid Approach	Accuracy	91.29%	Hybrid approach
[11]	UNSW-NB15	LR RF SVM	Accuracy	98.68%	Dataset imbalance Binary classification
[12]	UNSW-NB15	DNN CNN LSTM	Accuracy	99.76%	Dataset imbalance Binary classification
[13]	UNSW-NB15	dense DNN RNN	Accuracy	79.21%	Multi-class Classification Cross-validation
[14]	UNSW-NB15	CTGAN-with ML + DL	MCC	+13%	Multi-class Classification

TABLE I. shows summary of previous studies, in our research we will use The Training UNSW-NB15 which have 175,341 records and 45 features from UNSW research

repository, we will apply four deep leaning techniques on the dataset , first goal is enhance the quality of the dataset by solving class imbalance using SMOTE method and second goal is investigate the performance of four LSTM models which are: Long Short-Term Memory (LSTM), Bidirectional LSTM (BiLSTM) , Gated Recurrent Unit (GRU) and Bidirectional GRU (BiGRU), we use Binary classification to predict if the connection Normal or Attack, we will apply LSTM deep learning models on the UNSW-NB15 dataset. Then the performance results will be analyzed to identify the LSTM model that yields the highest performance for UNSW-NB15 dataset.

III.BACKGROUND

A. Long Short-Term Memory (LSTM)

Long Short-Term Memory (LSTM) is a specific architecture of RNN. The LSTM architecture consists of three main parts: the input gate that determines when the input data is significant enough to be remembered, and forget gate that decides whether to retain or discard the previous value. Lastly, the output gate for collecting and transmitting the stored value. One of the key advantages of LSTM networks is their ability to efficiently learn sequences of data [12].

B. Gate Recurrent Unit (GRU) :

The Gated Recurrent Unit (GRU) is a modified Architecture of the Long Short-Term Memory (LSTM) network, as proposed by Cho et al. It is designed to use fewer resources, which allows it to process data more quickly. The GRU consists of two main parts: the reset gate that decides whether to consider all or part of the previous value based on the state of the candidate layer, while the update gate decides how to adjust the current value[15].

C. Bidirectional LSTM (BiLSTM) and GRU (BiGRU) :

LSTM and GRU are one directional models that process data in one direction according to their timestamps. However, they cannot backtrack. In [16], researchers proposed the BiLSTM model, which is a both -directional data processing model. This allows the model to learn from data in both directions, resulting in more reliable system. Also the BiGRU model learn from data in both directions as BiLSTM

D. Dataset Description

In this research, we utilize the UNSW-NB15 training dataset, which was created by establishing at the University of New South Wales by Cyber Security Lab and published in 2015. The UNSW-NB15 network packets were generated using tool called the IXIA Perfect Storm in the Cyber Range Lab. This tool produces a hybrid of real modern normal activities and attack behaviors. The UNSW-NB15 training dataset comprises 175,341 records and includes 44 features categorized into nine different types of malicious activities: Denial of Service, Exploit, Analysis, worms, Backdoor, Shellcode, Fuzzers, Generic, and Reconnaissance, along with normal traffic. It is important to note that there is data imbalance within the UNSW-NB15 dataset, as the number of samples in some class is significantly higher than in classes in the dataset have and we binary classification with two classes ; one normal class and attack class, the Label 0 for normal samples and Label 1 for attack samples, the **Table II.**

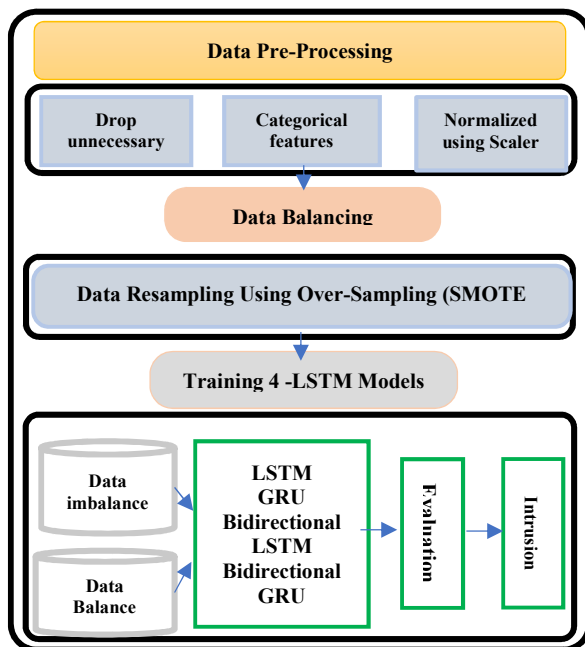
Shows the number of samples for each class. The dataset consists of 44 features grouped into six categories: content, time, and flow, basic, additional generated, and tagged. Specifically, features 1 to 30 represent integrated information gathered from data packets. Connection features are found from 31 to 37, while the characteristics numbered 38 to 42 are classified as general-purpose features. Finally, features 43 and 44 serve as labels.

TABLE II. THE NUMBER OF SAMPLES

Class Name	Number of samples
Attack Class label=1	119341
Normal Class label=0	56000

IV. METHODOLOGY

Fig. 1 shows the methodology is used in this research and structured as following steps:



A. Preprocess the Dataset:

UNSW-NB15 dataset contains 175,341 records and 44 features categorized with nine different types of malicious activities and the last feature 45 is class label. Many steps are taken for Preprocessing the Dataset; **Table III** shows the features initial and processed characteristics of before and after pre-processing:

- There are unnecessary features like id, so we drop these features: id, stcpb, dtcpb, attack_cat, state, proto and the current number of features is 39
- Preprocessing for categorical features and the service feature is preprocessing using One Hot Encoder method.
- Numerical features are normalized using Standard Scaler method.

TABLE III. DATASET CHARACTERISTICS AFTER PROCESSING

Detailed information	UNSW-NB15 Dataset
Domain	Iot Network
Number records	175,341
Initial features number	44
Final features number	39
Final features number after using one hot encoder	50
Number of instances Normal class	56000
Number instances Attack class	119341

B. Data Balancing:

Table III. Provides a detailed analysis of the imbalance ratio for binary classes in the dataset, it shows that the UNSW-NB15 before sampling. A widely used method for addressing imbalanced datasets and called resampling, which involves either reducing the size of the majority class that called under sampling or increasing the size of the minority class that called oversampling. Under-sampling involves the removal of samples from the major class, while over-samples involve the creation of new samples in the minor class [18]. SMOTE (Synthetic Minority Over-sampling Technique) is a popular method for addressing imbalanced datasets. SMOTE generate instances of the minority class, effectively increasing its representation to better match the majority class. The SMOTE method Preserves Information: Unlike under-sampling (removing majority class instances), SMOTE avoids potential loss of valuable information, and generates new data By interpolating between existing minority class instances, SMOTE creates synthetic samples, enhancing the dataset's diversity. But SMOTE Increased Computational Cost: Generating synthetic data can increase the size of the dataset, potentially leading to higher computational demands for model training and evaluation. In summary, SMOTE offers a valuable approach to handling imbalanced datasets by creating synthetic minority class instances, improving model performance while minimizing the risk of information loss and overfitting [18]. SMOTE can introduce noise and overfitting and synthetic data bias, potentially reducing precision. To overcome these limitations, several improved oversampling techniques have been developed. Like: Borderline SMOTE, Advanced SMOTE (A-SMOTE). These advanced methods refine the selection of minority samples used for synthetic generation, often based on the density of majority class samples within their k-nearest neighbors [19].

The second step is prepare the processed data from the first step to apply SMOTE method for solving data imbalances. The **Table IV.** Shows the results after oversampling.

TABLE IV. RESULTS AFTER OVERSAMPLING

Class Name	Number of samples BEFOR SMOTE	Number of samples AFTER SMOTE
Attack class label=1	119341	119341
Normal Class label=0	56000	119341

C. Performance Evaluation :

Evaluation metrics are used to measure the performance of model, we used confusion matrix to study the efficiency of each model. Confusion matrix contains information about the actual and predicted classes, the following Table VI. Shows the confusion matrix; Where the diagonal (TP and TN) refers to the set of samples that were predicted correctly, and the diagonal (FN and FP) refers to the set of samples that were predicted wrong [17]. Intrusion Detection can be regarded as a binary classification with the predicted binary label (0,1), the Label 0 for normal connection and Label 1 for attack records, Researchers used a confusion matrix to present the classification results, the definitions confusion matrix for Intrusion Prediction are shown in **Table VI**.

TABLE VI. CONFUSION MATRIX FOR INTRUSION DETECTION

Predicated	Actual	
	Attack	Normal
Attack class label=1	TP	FN
Normal Class label=0	FP	TN

Based on the confusion matrix, we calculated the performance metrics of the classification models. Metrics can be derived from the confusion matrix; Accuracy, precision, recall, f-score are used to evaluate each models, the following equation, show each metric:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

$$F1 - score = \frac{2(Precision * recall)}{Precision + recall} \quad (4)$$

$$AUC = \int TPR(t)[-FPR'(t)]dt \quad (5)$$

Where TPR is the true positive rate ($TPR = \frac{TP}{TP + FN}$), FPR is the false positive rate ($FPR = \frac{FP}{FP + TN}$), and t is the classification threshold. The range of the AUC was [0.5, 1], with larger AUC values representing better classification performance [20].

In our experiments we use python collab for creating prediction models. The experiments will be conducted on personal computer with a 2.4 GHz core i3 CPU and 8GB RAM under windows 11. First step in the experiments' setup is to load the processed and the categorical attributes were encoded as one-hot encoder method using scikit-learn libraries, then the attributes are normalized using Standard Scaler method. The second step is prepare the processed data from the first step to apply SMOTE method for solving data imbalances. **Table IV** shows the results after oversampling. The Third step is to prepare the data for applying the model on it, in this step the data will be partitioned into two subsets, the first subset will be used for training the model, and the second one for testing the well-trained model. Empirical studies show that the best results are obtained if we use 20-30% of the data for testing, and the remaining 70-80% of the data for training [21], so we split the data into 80% for

training and 20% for testing, then the LSTM deep learning models is defined. We have two experiments is applied , first experiment for training the dataset by four types LSTM models without SMOTE over- sampling method , the second experiment training the dataset by four types LSTM models with SMOTE over sampling method. For creating deep learning models, we import keras with tensor flow libraries using Python, the keras library can work with python libraries like scikit-learn library. The sequential LSTM type is used with one hidden layer and the number of hidden neurons is 64 and the RLU the activation function for each hidden neuron, and we use ADAM function as optimizer function and for the output we use sigmoid functions. The **Table VII**. Shows the parameters and hyper parameters are tuning when building the models.

TABLE VII. TUNING THE PARAMETERS

Parameters	Models (LSTM,BiLSTM,GRU,BiGRU)
Activation function	RLU
Optimizer	ADAM
Loss function	Binary Cross Entropy
Batch-size	16
Number of Epochs	10
Number of Hidden Neurons	64
Hidden layers	1

The **Table VII**. shows that after training with additional layers, we conducted many experiments with vary hidden layers and hidden neurons and find the best results when have one hidden layers, after defining the LSTM Models then training data set is trained by the desired model, and the last step is to evaluate the performance of the model by applying the model on the testing test; the result of the evaluation is in the confusion matrix where the accuracy derived from the confusion matrix will be studied

V.RESULT AND DISCUSSTION

After implementing the LSTM models UNSW-NB15 Dataset we evaluate the performance efficiency of each model using confusion matrix; to calculate the accuracy of the model; and be able to compare between four models; and find the model with the best accuracy among of them.

A. Training the Dataset without SMOTE over- sampling:

Following **Table VIII**. Shows the result on the NSW-NB15 Dataset. The results shows the BiLSTM model has better performance with (0.9468) Accuracy and (0.9801) AUC, the results show the BiLSTM outperforms the other models are used.

TABLE VIII. THE EVALUATIONS RESULTS WITHOUT SMOTE

Metrics	Models			
	LSTM	BiLSTM	GRU	BiGRU
AUC	0.9790	0.9801	0.9712	0.9747
Accuracy	0.9455	0.9468	0.9446	0.9453
Precision	0.95	0.95	0.95	0.94
Recall	0.92	0.93	0.92	0.93
F1-score	0.93	0.94	0.94	0.93

B.Training the Dataset with SMOTE over- sampling:

The following **Table IX.** Shows the result on the NSW-NB15 Dataset and shows the SMOTE Training time and Memory usage. The BiLSTM has better performance with **(0.95)** Accuracy and **(0.9953)** AUC the results show the BiLSTM model outperforms the other models.

TABLE IX. THE EVALUATIONS RESULTS WITH SMOTE

Metrics	Models			
	LSTM	BiLSTM	GRU	BiGRU
AUC	0.9928	0.9953	0.9910	0.9923
Accuracy	0.9470	0.9501	0.9450	0.9455
Precision	0.94	0.95	0.94	0.94
Recall	0.94	0.95	0.94	0.94
F1-score	0.94	0.95	0.94	0.94
SMOTE Training Time	7 Minutes	11 Minutes	9 Minutes	11 Minutes
Memory usage	2.1 GB	2.2 GB	2.2 GB	2.2 GB

VI.CONCLUSION

This research investigates the performance of deep learning models, specifically Recurrent Neural Networks (RNNs), to enhance Intrusion Detection in Internet of Things (IoT) systems for identify and predict if the connection is normal or attack. The IoT, characterized by interconnected devices, presents unique security challenges due to its inherent vulnerabilities. Traditional security measures, such as firewalls and signature-based Intrusion Detection Systems (IDSs), have limitations in detecting sophisticated attacks. To address these limitations, this study explores the performance of deep learning algorithms, including Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), Bidirectional LSTM, and Bidirectional GRU, in developing a robust intrusion detection model. These models are designed to effectively analyze network traffic data and identify and predict if the connection is normal or attack. The research utilizes the UNSW-NB15 dataset, which serves as a benchmark for intrusion detection research and incorporates real-world network traffic along with synthetic attack scenarios. To tackle the class imbalance present in the dataset, the study employs the Synthetic Minority Over-sampling Technique (SMOTE) to improve model performance. The performance of the four LSTM models is evaluated both with and without SMOTE, aiming to identify the most effective approach for accurate intrusion detection systems in IoT networks. The models are compared using metrics such as Accuracy, Precision, Recall and F1-score. The results indicate that the model with the best metrics is the Bidirectional LSTM (BiLSTM), achieving an Accuracy of 0.9468 , AUC of 0.9801 without SMOTE and an Accuracy of 0.9501, AUC of 0.9953 with SMOTE. To validate the differences are meaningful, the t-test [22] is conducted and we obtained value is less than 0.05 that means there is a

significant difference between two experiments, after analyzing the results, it can be concluded that the BiLSTM model outperformed the other models. Furthermore, using the SMOTE method for balancing the data improved the results.

This research significantly enhances IoT intrusion detection by employing Bidirectional LSTM (BiLSTM), leveraging BiLSTM's ability to analyze sequential network traffic data and identify complex attack patterns, the research achieves improved accuracy and performance, as evidenced by high accuracy and AUC scores. Furthermore, the use of the Synthetic Minority Over-sampling Technique (SMOTE) effectively addresses class imbalance, further boosting the model's ability to detect attacks. Statistical validation through a t-test confirms the significance of these improvements, demonstrating that BiLSTM with SMOTE provides a more robust and adaptable solution for securing vulnerable IoT environments against sophisticated threats. The authors in [11] , [12] obtained 98.68% , 99.75 % accuracy using ML and LSTM respectively , but the authors haven't handling imbalanced datasets so may yield unsatisfactory results and overfitting, but in our research we handling imbalanced datasets and obtained reliable results, the authors in [13] obtained 79.21% accuracy using LSTM with Multi-class Classification, but in our research we use Binary Classifications.

The findings of this research will contribute to the development of more secure IoT systems by providing insights into the effectiveness of BiLSTM-based intrusion detection algorithms using the SMOTE method.

In future work for obtaining high and reliable results, we will investigate different techniques for handling imbalance Dataset.

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